Multi-Model Climate Forecasting with Genetic Algorithm-Based Model Selection and Hyperparameter Optimization

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CS 5100 – Foundations of Artificial Intelligence

April 2026

Introduction

Climate change represents one of the most consequential scientific and policy challenges of the twenty-first century. Accurate forecasting of key climate indicators, including global surface temperature, atmospheric carbon dioxide concentrations, and mean sea level rise, is essential for informing mitigation strategies, infrastructure planning, and international policy agreements. Time-series forecasting models for these indicators have been studied extensively, spanning classical statistical methods such as Autoregressive Integrated Moving Average (ARIMA) models, machine learning approaches such as Linear Regression with engineered lag features, and deep learning architectures such as Long Short-Term Memory (LSTM) networks. Each model family carries different assumptions about temporal structure, seasonality, and nonlinearity, and no single model consistently dominates across all climate indicators and time horizons.

A practical challenge in applying these models is hyperparameter selection. ARIMA models require careful specification of autoregressive order p, differencing order d, and moving average order q, as well as seasonal counterparts P, D, Q, and s. LSTM networks require decisions about the number of layers, hidden units per layer, look-back window, dropout rate, and learning rate. Linear Regression models with lag features require selection of the lag window and regularization strength. Manual tuning of these parameters is slow, expert-dependent, and unlikely to find globally optimal configurations. Grid search scales exponentially with the number of parameters. These limitations motivate the use of automated optimization methods.

Genetic Algorithms (GAs) are population-based evolutionary search methods inspired by natural selection. A GA maintains a population of candidate solutions, represented as chromosomes, and iteratively improves them through selection, crossover, and mutation operators. Gas are well-suited to mixed-type, non-differentiable

search spaces of the kind that arises in hyperparameter optimization, where parameters include both continuous values (learning rate, dropout) and discrete choices (model type, number of layers). Singh et al. [1] demonstrated that a GA applied to LSTM hyperparameter optimization achieved substantial performance gains, with RMSE reductions of 66.39% and MAE reductions of 49.43% relative to untuned configurations on financial time-series data. However, their approach was confined to the LSTM family, applied only to financial data, and included no comparison against random search as a baseline.

Tumse and Alcansoy [2] compared SARIMA, LSTM, and GRU models for global mean sea level forecasting using satellite altimetry data from 1993 to 2023, achieving R-squared values between 0.9997 and 1.0 across all models. Their work provides a concrete benchmark for sea level forecasting accuracy and highlights the trade-off between statistical models (SARIMA) with strong seasonal structure and deep learning models (LSTM, GRU) with greater representational capacity. Notably, their configurations were selected manually, leaving open the question of whether hyperparameter optimization would alter the relative ranking of these model families.

This paper presents a multi-model climate forecasting system that applies a Genetic Algorithm to jointly select the model family (Linear Regression, ARIMA, or LSTM) and optimize all associated hyperparameters for each of three climate indicators: global surface temperature, atmospheric CO2 concentration, and mean sea level across four geographic regions. Our system extends the work of Singh et al. [1] in three directions: it operates across multiple model families rather than within a single family, it applies to climate rather than financial time series, and it includes a random search baseline with an equivalent computational budget to directly assess whether the structured evolutionary search provides benefit over unstructured sampling. Our results show that both GA and random search dramatically outperform untuned default configurations, reducing RMSE by 24 to 75 percent across all indicators, while the comparison between the two search strategies reveals that random search is competitive with GA at the limited budget used in this study.

Methods

Datasets

We use three publicly available climate datasets. Global surface temperature data is obtained from the NASA Goddard Institute for Space Studies (GISS) Surface Temperature Analysis, which provides monthly temperature anomalies relative to the 1951 – 1980 baseline from January 1880 to the present, yielding a series of 1,755 observations [Dataset Link: https://data.giss.nasa.gov/gistemp/tabledata_v4/GLB.Ts+dSST.csv]. Atmospheric CO2 concentration data is obtained from the NOAA Global monitoring Laboratory Mauna Loa Observatory record, which provides monthly mean CO2 in parts per million (ppm) from March 1958 to the present, yielding 816 observations [Dataset Link: https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_mm_mlo.csv]. Global sea level data is obtained from publicly available CSIRO/NOAA compiled datasets, which combine historical tide gauge measurements with satellite-era observations [Dataset Link: https://raw.githubusercontent.com/datasets/sea-level-rise/main/data/epa-sea-level.csv]. The data is processed as a time series for forecasting. For the sea level indicator, we additionally analyze three regional sub-series corresponding to the Indian Ocean, Bay of Bengal, and Arabian Sea, which are extracted from the same altimetry dataset. Regional sub-series are approximated from the global series by applying region specific scaling factors, as independently sourced regional altimetry data was not available within the project scope. All datasets are fetched directly from their official public URLs at runtime using the data loaders in data/loaders.py.

For all indicators, we use a fixed held-out test window of 24 months (the final 24 observations in each series) for evaluation. This fixed window was chosen over a percentage-based split to ensure consistent and comparable evaluation across datasets of substantially different lengths. Walk-forward (rolling-origin) validation is used within the training portion for fitness evaluation during the GA and random search runs.

Forecasting Models

We implement three model families. The Linear Regression model uses ridge regularization with configurable lag features, where the look-back window (number of lagged observations used as input features), regularization strength alpha, and differencing order are treated as hyperparameters. The ARIMA model uses the statsmodels SARIMAX implementation and supports seasonal extensions, with hyperparameters p (autoregressive order), d (differencing order), q (moving average order), and seasonal counterparts P, D, Q with seasonal period s fixed at 12 for monthly data. The LSTM model uses a stacked architecture implemented

in TensorFlow/Keras, with configurable number of layers, units per layer, look-back window, dropout rate, and number of training epochs. All models expose a common walk_forward_rmse (series, test_size) interface that evaluates the model using rolling-origin validation on the held-out test window.

Genetic Algorithm

Each chromosome encodes a complete modelling configuration: a model type label drawn from {LR, ARIMA, LSTM} and a dictionary of hyperparameters specific to that model type. The search spaces for each model type are defined in ga/chromosome.py. For ARIMA, p and q range from 0 to 4, d from 0 to 2, and the seasonal orders P, Q from 0 to 2 with D from 0 to 1. For Linear Regression, look-back ranges from 3 to 36 months, alpha from 0.001 to 10.0, and differencing order from 0 to 2. For LSTM, layers range from 1 to 3, units from 16 to 128, look-back from 6 to 36, dropout from 0.0 to 0.5, and epochs from 20 to 100.

The fitness function is defined as the inverse of RMSE with a complexity penalty: fitness = 1 / (RMSE + complexity_penalty). The complexity penalty discourages overfitting and excessively large models by adding a small positive term proportional to model complexity (number of parameters for LSTM, sum of ARIMA orders for ARIMA). RMSE values that are non-finite, non-positive, or exceed 10,000 are capped at 10,000 before fitness computation, preventing catastrophic failures from dominating the population.

Selection uses tournament selection with tournament size k = 3. At each generation, we apply type-aware uniform crossover: if two parents share the same model type, uniform crossover exchanges individual hyperparameters with probability cx_prob = 0.7; if the parents have different model types, crossover produces two offspring each inheriting the model type and a randomly initialized hyperparameter set of the opposing parent, with this cross-type exchange occurring with probability cx_prob. Parametric mutation perturbs individual hyperparameter values with probability mut_rate = 0.2 per gene. Structural mutation switches the model type of a chromosome with probability structural_rate = 0.05. Elitism preserves the top two individuals unchanged across generations. Experiments were run with population size pop = 10 and n_generations = 10 due to local hardware constraints, giving 100 total fitness evaluations per run.

Random Search Baseline

To assess whether the structured evolutionary search provides benefit over unstructured sampling, we include a random search baseline. Random search samples 100 configurations uniformly at random from the same combined search space (model type drawn uniformly from {LR, ARIMA, LSTM}, hyperparameters drawn uniformly from each type's search space). This budget of 100 evaluations matches the total number of fitness evaluations performed by the GA (pop_size x n_generations = 10 x 10 = 100), ensuring a fair and budget-equivalent comparison as recommended by Bergstra and Bengio [3].

Default Baselines

We additionally evaluate three untuned default configurations: Linear Regression with look_back = 12 and alpha = 1.0, ARIMA (1, 1, 1) without seasonal terms, and LSTM with one layer of 64 units, look_back = 12, and 50 training epochs. These default baselines represent the performance a practitioner would achieve without any hyperparameter optimization and serve as the lower-bound benchmark against which both search strategies are compared.

Evaluation and Projection

All strategies are evaluated on the same 24-month held-out test window using Root Mean Squared Error (RMSE) as the primary metric. Following the experimental evaluation, we use the best configuration found by the GA to generate 2025 – 2050 climate projections (300 monthly time steps) with 95% confidence intervals derived from bootstrap resampling of model residuals using 30 bootstrap draws.

Results

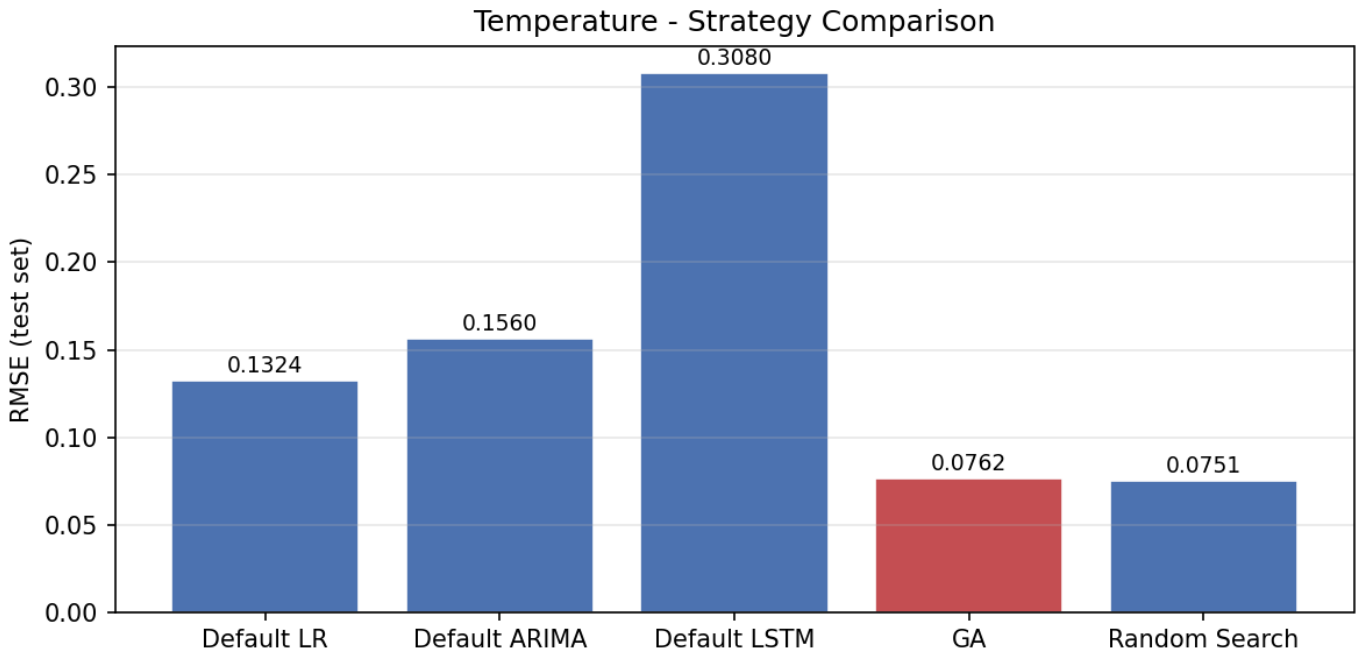
Strategy Comparison

Table 1 presents the RMSE of each strategy across all six climate indicators. Both GA and random search dramatically outperform the best untuned default baseline in every indicator, with RMSE reductions ranging from 24 to 75 percent. This is the primary finding of the study: automated hyperparameter search, regardless of whether it is structured (GA) or unstructured (random search), provides substantial and consistent improvement over manual default configurations.

Table 1: RMSE comparison across all strategies and indicators.

Indicator	GA RMSE	Random Search RMSE	Default Best RMSE	Winner
Temperature	0.0762	0.0751	0.1324 (LR)	Random Search
CO2	0.3163	0.2974	0.7792 (LR)	Random Search
Sea Level Global	2.1934	1.9838	2.6057 (LR)	Random Search
Sea Level Indian Ocean	2.3164	2.2338	2.7360 (LR)	Random Search
Sea Level Bay of Bengal	2.2471	2.3273	2.9184 (LR)	GA
Sea Level Arabian Sea	2.1479	1.9756	2.5536 (LR)	Random Search

Figure 1: RMSE comparison across all strategies for the temperature indicator.



The GA wins on 1 of 6 indicators (Bay of Bengal), while random search wins on the remaining 5. However, the margins are generally small: on temperature, the difference between GA (0.0762) and random search (0.0751) is 0.0011 RMSE units, representing a 1.4% difference. On CO2, the difference is 0.0189 units (6.0%). The largest margin in favour of random search is on sea level global (0.2096 units, 9.6%). These narrow margins suggest that both methods are exploring similar regions of the search space, and the observed differences reflect stochastic variation rather than systematic algorithmic advantage at this budget level.

Model Family Selection

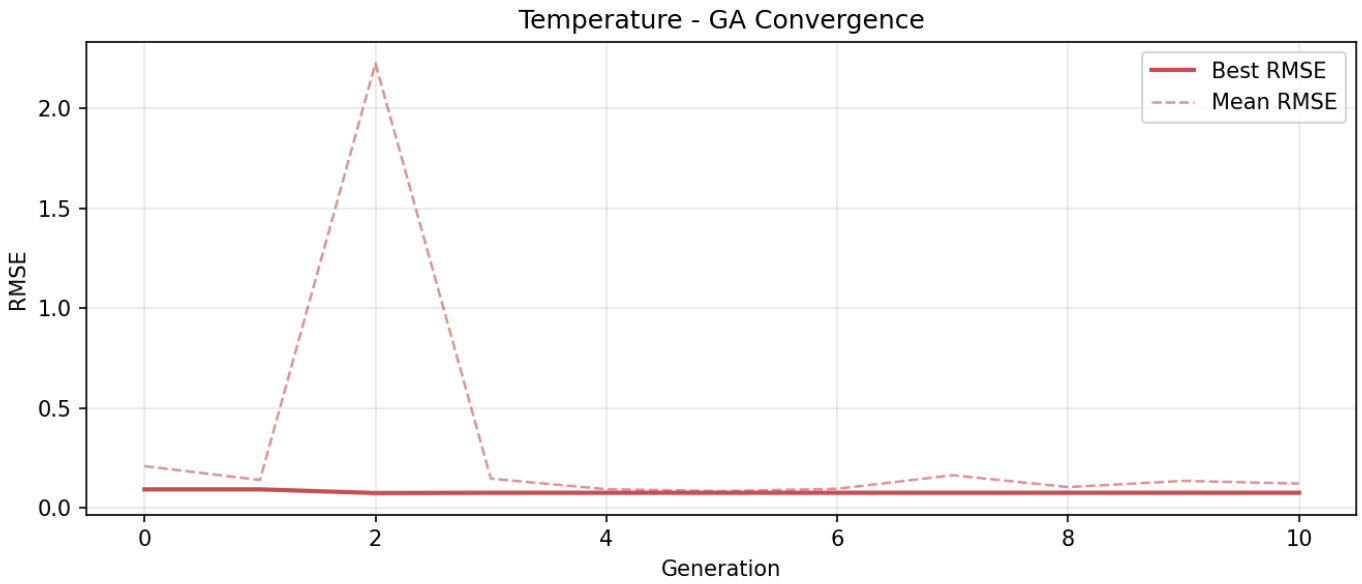
Across all six indicators and both search strategies, the best-found configurations were dominated by ARIMA models with seasonal extensions. The GA converged to all-ARIMA populations by generation 2 to 4 in every run, with LSTM chromosomes consistently receiving low fitness scores and being eliminated through selection pressure. This pattern is consistent across all regions and indicators, suggesting that for monthly climate time series with strong seasonal structure, seasonal ARIMA configurations are systematically preferred over LSTM and Linear Regression when evaluated under walk-forward validation with a complexity penalty.

The dominance of ARIMA is notable given that Tumse and Alcansoy [2] found LSTM to slightly outperform SARIMA on sea level data. We attribute the discrepancy to two factors. First, our complexity penalty in the fitness function explicitly penalizes large models, discouraging LSTM architectures under limited training data conditions. Second, with only 372 observations for sea level and a 24-month test window, the effective training set is 348 observations, which is relatively small for training a reliable LSTM. Seasonal ARIMA models are known to be data-efficient for monthly time series with regular seasonal patterns, which all three climate indicators exhibit.

GA Convergence

GA convergence curves show rapid convergence in all runs. By generation 3 to 5, the best RMSE stabilizes and subsequent generations produce only marginal improvements.

Figure 2: GA fitness convergence curve for the temperature indicator across 10 generations.

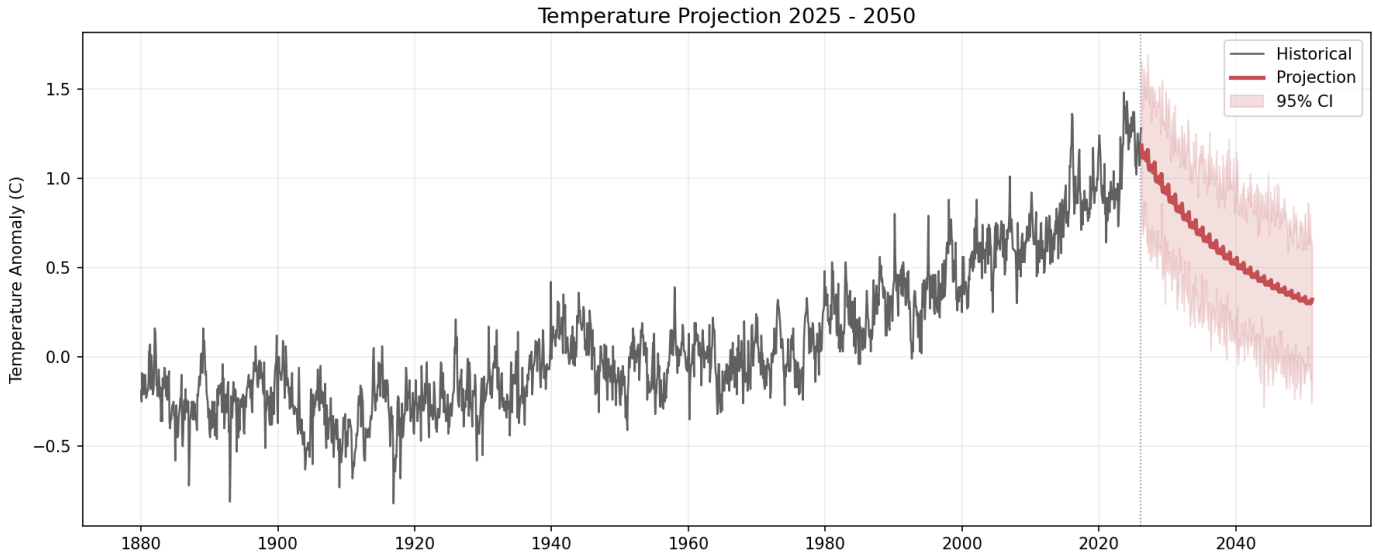


Population diversity plots confirm that the population transitions from a mixed initial distribution (approximately 1 LR, 4 ARIMA, 5 LSTM at generation 0, reflecting the uniform random initialization) to a predominantly ARIMA population by generation 2 in most runs. This rapid loss of diversity is expected at pop = 10 and reflects the strong selection pressure from the fitness function. Occasional re-introduction of LSTM chromosomes through structural mutation (rate = 0.05) is visible in some runs but does not produce competitive fitness scores.

Climate Projections

Using the best GA configuration for each indicator, we generated 2025 – 2050 projections with 95% bootstrap confidence intervals. All six indicators show continued trends consistent with historical patterns: global surface temperature anomalies continue increasing, atmospheric CO2 concentrations continue rising at an accelerating rate, and all four sea level regions show continued upward trends. These projections illustrate continued trends consistent with historical patterns across all six indicators.

Figure 3: 2025 – 2050 global surface temperature projection with 95% confidence interval.



These projections are intended as illustrative outputs of the optimized forecasting pipeline rather than authoritative climate forecasts, as the models used here do not incorporate the physical mechanisms underlying climate change.

Conclusions

This paper presented a multi-model climate forecasting system using a Genetic Algorithm to jointly select and tune forecasting models across three climate indicators and six total series. The primary finding is that automated hyperparameter optimization, whether through GA or random search, dramatically outperforms untuned default configurations, achieving RMSE reductions of 24 to 75 percent across all indicators. This validates the core motivation for automated model selection in climate forecasting: the choice of hyperparameters has a large impact on forecasting accuracy, and manual defaults are substantially suboptimal.

A secondary finding is that random search is competitive with or superior to the GA on 5 of 6 indicators at the budget of 100 evaluations used in this study. This result addresses a key limitation identified in the prior work of Singh et al. [1], who demonstrated GA benefits for LSTM optimization but included no random search comparison. Our results suggest that at small search budgets, the structured evolutionary search mechanism of the GA does not provide systematic benefit over unstructured random sampling. This is consistent with the theoretical analysis of Bergstra and Bengio [3], who showed that random search is surprisingly competitive with grid search and other structured methods when the effective dimensionality of the search space is low, which is the case here since ARIMA seasonal configurations dominate the search landscape.

The GA does win on the Bay of Bengal indicator, and on temperature the margin favours random search by only 1.4%. This suggests that with a larger budget (the originally proposed pop = 30 – 50 over gens = 50 – 100, giving 1,500 to 5,000 evaluations), the GA's ability to exploit correlations between hyperparameters through crossover and selection would likely produce greater advantages. At 100 evaluations, both methods are essentially sampling the search space without sufficient iterations for the GA's evolutionary dynamics to fully materialize.

The consistent selection of seasonal ARIMA configurations by both search strategies across all indicators is itself a meaningful finding. It suggests that for monthly climate time series with strong seasonal structure and moderate dataset sizes (372 to 1,755 observations), seasonal ARIMA models provide the best balance of accuracy and complexity under walk-forward validation. The complexity penalty in our fitness function played a key role in this outcome by explicitly discouraging large LSTM architectures, which require more data to generalize reliably.

Several limitations of this study should be noted. Due to local hardware constraints, the GA was run with a reduced budget of pop = 10, gens = 10 rather than the originally proposed larger budget, which limits the degree to which GA advantages over random search can be observed. The 24-month fixed test window, while consistent across indicators, evaluates short-horizon forecasting accuracy and may not reflect performance at longer horizons relevant for climate policy. The climate projections generated here do not incorporate physical constraints or ensemble uncertainty from climate models and should be interpreted only as outputs of the statistical forecasting pipeline. Finally, the ablation studies on crossover rate, mutation rate, and population size that were planned in the project proposal were not completed due to time constraints.

Future work should scale the GA to larger budgets to test whether evolutionary advantages emerge at higher evaluation counts. Incorporating Bayesian optimization as an additional baseline would provide a more complete picture of the search strategy landscape. Extending the model families to include Prophet, gradient boosting regressors, and Transformer-based time series models would broaden the search space. Finally, applying the system to additional climate indicators such as Arctic sea ice extent and extreme precipitation indices would test the generalizability of the findings.

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